

Tushar Das

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 Portfolio |  LinkedIn |  GitHub

Education

Shahjalal University of Science and Technology (SUST)

B.Sc. Engineering in Software Engineering, Institute of ICT (IICT)

CGPA: 3.83/4.00 | Credits Completed: 124/163

Sylhet, Bangladesh

4th Year, 1st Semester

Research Experience

Undergraduate Research Assistant — AI-Driven Data Analytics for Bangladesh's Primary Education Jan 2026 – Jul 2026

- Contributed to "Transforming Bangladesh's Primary Education System through AI-Driven Data Analytics," under Fazle Rabbi Rakib and Mafuzur Rahman Emon (IICT, SUST).

Co-Author — An Ensemble Learning Framework for Crop Recommendation with LLM-Based Explanations Dec 202 – Aug 2026 Submitted – ECCE2027

- Developed a 99.55% accurate crop recommendation ensemble optimized using Optuna and cross-validated with SHAP and LIME.
- Deployed a local Qwen2.5 LLM to translate complex AI attributions into plain-language explanations for farmers.

Technical Skills

- Languages:** Java, C, C++, JavaScript, Python, SQL, Dart
- Backend:** Spring Boot, Node.js, Express.js, RESTful APIs, JWT Authentication, RBAC
- Frontend & Mobile:** React.js, Flutter, HTML5, CSS3
- Data Science & ML:** NumPy, Pandas, Tensorflow, Keras
- Databases:** MySQL (Relational Schema Design, ER Diagrams)
- Concepts:** Object-Oriented Programming (OOP), SDLC, SRS Documentation

Projects

SUST Medical Center Management System

Node.js, Express, MySQL, React.js

- Led full SDLC for a real enterprise system, from direct requirement elicitation with SUST Medical Center staff to deployment.
- Designed a normalized MySQL schema with defined entity relationships and key constraints.
- Built a multi-role REST API (patient, doctor, nurse, admin) with custom business logic and authorization; developed the React.js frontend for staff and doctor workflows.

Profile Based Task Management System

Java, Spring Boot, MySQL, JWT

- Implemented username/password authentication, JWT session handling, email verification and Role-Based Access Control (RBAC) to govern permissions between user and admin level.

CampusConnect - Social Media Platform for SUST Students

Flutter, Dart, Firebase

- Led design and development of a cross-platform mobile app that enables intra-campus communication for the SUST community.

Full-Stack Hall Management System

Java, Spring Boot, React, MySQL

- Developed a residential hall management system with a scalable RESTful backend (Spring Boot) and a React frontend.

Snake Game

C, SDL

- Built a graphics-based Snake Game using the SDL framework, applying procedural programming and data structures.

Activities & leadership

- Sports Secretary, SWE Society** — Elected leader organizing inter-departmental and intra-university sports tournaments; managed event logistics, team coordination, and budget allocation.

Achievements

- Finalist, NextGen Hackathon — IIUC CSE Fest 2025